Software Development Life Cycle (SDLC) Approach

Project: E-commerce Platform for Electronic Gadgets

1. Recommended Approach

For the development of the e-commerce platform, the **Agile methodology** is recommended. Agile is suitable for this project because it allows flexibility, continuous feedback, and incremental delivery, which are essential for building a customer-facing platform where user experience is critical.

2. Key Phases of the SDLC

1. Requirement Analysis

- Gather detailed functional and non-functional requirements from stakeholders.
- Document requirements in BRD and FRD.
- Identify user journeys and prepare use cases.

2. Design Phase

- Prepare wireframes and UI/UX prototypes for approval.
- Define the system architecture, database schema, and class diagrams.
- Review designs with stakeholders before moving to development.

3. Development Phase

- Implement features in short, iterative sprints.
- Prioritize high-value features first (user registration, product catalog, checkout).
- Ensure code is modular and scalable for future enhancements.

4. Testing Phase

- Conduct unit testing, integration testing, and system testing.
- Perform user acceptance testing (UAT) to validate requirements.
- Ensure security and performance testing for reliability.

5. Deployment Phase

- Deploy features incrementally to a staging environment.
- Roll out the application to production once stable.
- Provide training and documentation to the client team.

6. Maintenance & Monitoring

- Monitor platform performance and user feedback post-launch.
- Fix bugs, release patches, and add enhancements through future sprints.

3. Benefits of Using Agile

- **Flexibility:** Requirements can be refined based on ongoing client feedback.
- **Early Value Delivery:** Core features are released early, allowing customers to start using the platform sooner.
- **Collaboration:** Regular sprint reviews keep all stakeholders aligned.
- **Risk Reduction:** Continuous testing and integration reduce the chance of failures at the end.

Agile Events

1. **Sprint Planning** – The team collaborates with stakeholders to define the scope and goals for the upcoming sprint.
2. **Daily Stand-ups** – Short daily meetings where team members share progress, challenges, and plans for the day.
3. **Sprint Review** – Conducted at the end of each sprint to demonstrate completed features to stakeholders and gather feedback.
4. **Sprint Retrospective** – A reflection session where the team discusses what went well, what didn't, and how processes can be improved for the next sprint.